Signal Processing: Image Communication 80 (2020) 115648

Contents lists available at ScienceDirect

Signal Processing: Image Communication

journal homepage: www.elsevier.com/locate/image

Visual question answering model based on visual relationship detection ☆

Yuling Xi [a], Yanning Zhang [a], Songtao Ding [b], Shaohua Wan [c,d,1,*]

[a] School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China
[b] School of Computer Science & Technology, Xi'an University of Posts & Telecommunications, Xi'an 710121, China
[c] School of Information and Safety Engineering, Zhongnan University of Economics and Law, Wuhan 430073, China
[d] State Key Laboratory for Novel Software Technology, Nanjing University, Nanjing 210023, China

ARTICLE INFO

Keywords:
Visual question answering
Appearance features
Relationship predicate
Word vector similarity

ABSTRACT

visual question answering (VQA) is a learning task involving two major fields of computer vision and natural language processing. The development of deep learning technology has contributed to the advancement of this research area. Although the research on the question answering model has made great progress, the low accuracy of the VQA model is mainly because the current question answering model structure is relatively simple, the attention mechanism of model is deviated from human attention and lacks a higher level of logical reasoning ability. In response to the above problems, we propose a VQA model based on multi-objective visual relationship detection. Firstly, the appearance feature is used to replace the image features from the original object, and the appearance model is extended by the principle of word vector similarity. The appearance features and relationship predicates are then fed into the word vector space and represented by a fixed length vector. Finally, through the concatenation of elements between the image feature and the question vector are fed into the classifier to generate an output answer. Our method is benchmarked on the DQAUAR data set, and evaluated by the Acc WUPS@0.0 and WUPS@0.9.

1. Introduction

In recent years, with the development of artificial intelligence technology, methods based on deep learning have achieved great success in visual perception tasks such as target recognition, image classification, and pose estimation [1–3]. However, in complex visual tasks, there is still a big gap between machine vision model and human beings for a lack of capabilities of deeper image understanding and target relationship reasoning. The visual question answering (VQA) combines various artificial intelligence technologies such as machine vision, natural language processing, and relational reasoning, and thus is capable of cross-modal understanding (image understanding and semantic generation). Therefore, VQA system helps narrow the gap between machine and human beings in terms of understanding of visual scenes.

VQA is a task system that takes a image and a open-ended natural language question about the image as input and takes generate a natural language answer as output. As VQA mainly involves machine vision and natural language processing, therefore, combining the machine vision algorithm with the natural language processing algorithm to construct the combined model becomes the mainstream method to solve VQA. This combined structure first uses a deep learning architecture to extract visual features, and then uses a Recurrent Neural Network (RNN) method to generate image text descriptions. Ma et al. [4] used three CNNs to complete the VQA. The first CNN is responsible for encoding the image features; the second CNN is responsible for each word that comprises the question sentence; and the third CNN uses the multimodal convolution layer to learn their joint representation to classify the candidate answers. Gao et al. [5] used a more complex model structure that consists of four parts: a Long Short Term Memory (LSTM) network for extracting question expressions; a Convolutional Neural Network (CNN) for extracting visual features; an LSTM that stores the language context in the answer; and a fusion component responsible for combining the information from the first three parts and generating an answer. Malinowski et al. [6] drew on recent advances in natural language processing and computer vision and proposed a method for automatically answering image questions. Ren et al. [7] suggested using neural networks and visual semantic embedding to replace the pre-processing processes such as target detection and image segmentation for answer prediction, and achieved good results on public benchmark data sets.

☆ No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.image.2019.115648.

* Corresponding author.
E-mail address: shaohua.wan@ieee.org (S. Wan).

[1] Senior Member, IEEE.

https://doi.org/10.1016/j.image.2019.115648
Received 22 May 2019; Received in revised form 26 August 2019; Accepted 17 September 2019
Available online 27 September 2019
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Although the research on VQA model has made great progress, the accuracy of the model evaluation is not high. When the model deals with complex images, the accuracy of the model is low, and even the complex model proposed in [8] has a passing rate of only 60% in the Turing test. This is mainly because the structure of the current VQA model is relatively simple, and the answer is simple in terms of both content and form. At the same time, the model requirement for the accuracy of the image target detection is very high. Therefore, when dealing with images with complex background and more prior knowledge is needed for logical reasoning, it is difficult to give an accurate answer. A paradigm analysis of the model that makes a wrong answer is given in [8]. Through analysis, it is found that these errors are mainly common sense errors of picture content, model attention and human attention deviation, lack of higher-level logical reasoning ability, and lack of target type and relationship structure in training library

To solve the above problems of VQA model, a VQA model (see Fig. 1) based on multi-objective visual relationship detection is proposed in this paper, which is inspired by the research on the target relationship in the image [9]. Firstly, the target relationship detection model is pre-trained, and then the appearance relationship feature is used to replace the image features extracted from the original target. At the same time, the appearance model is extended by the word vector similarity principle of the relation predicate, and the appearance features and relationship predicates are sent to the word vector space and are represented by fixed-size vectors. Finally, the integrated vector is sent to the classifier to generate an answer output, through the cascading of elements between the picture feature vector and the question vector. Our contributions consists of three main components:

(1) Judgment of the interrelations between the objects is added in traditional VQA visual tasks, because it is found through statistics such problems cover a larger proportion.
(2) The principle of word vector similarity is introduced in judgment of interrelations, transforming the problem of target relation into the problem of distance calculation of word vector space.
(3) Attention mechanism guided by problematic words is added to guide the attention to specific image regions.

2. Related work

Detection of target relation includes additionally training related with target relation based on the original VQA model. Before, it extracts only the target characteristics, but besides extracting target characteristics, it also extracts the relations between the targets as the reserves. Moreover, target appearance relation model of relation predicates is brought in for extra training. In case the problem is correlated with the relations between and the targets, the relation predicates between the targets would be input as the answer when answering questions. As our approach involves different areas such as machine learning, target detection, target relationship detection, and natural language processing, we introduce related work as follows:

Data set: The commonly used VQA database is [10]. The images in the data set usually come from the Flickr data set [11] and the MS COCO data set [12], and the data set is combined with human questions and basic answers to form a VQA data set. As the VQA model continues to improve, the rote learning strategy based on the language prior knowledge and example question/answer gains scores in the algorithm evaluation. However, the rote learning strategy actually hinders the effective evaluation and comparison of the VQA algorithms. Therefore, Goyal et al. [13] proposed an improved VQA v2 data set. This new database mainly copes with the imbalanced number of questions and answers in the VQA. Each question is associated with two images. The association is for ensuring that there is a peculiar answer to each image, avoiding random answers.

Target relationship detection: Visual relationship detection involves detecting targets in an image and understanding the interrelationship between these targets. Galleguillos et al. [14] proposed a context-based object classification model that is capable of combining different types of context information in computer vision for object classification. Choi et al. [15] introduced a new data set containing instances of different object classes and proposed an effective model to improve scene classification and object detection. Gould et al. [16] proposed a method for image segmentation using spatial relationships between objects. They defined four spatial relationships: above the target, below the target, inside the target, and around the target. Although the spatial relationship between the targets is important, they did not consider non-spatial relationships such as pull (action) and ratio (comparison). Some researchers collected examples of various target relationships from images [17–19] and videos [20,21], and they helped the model map these relationships from images to natural language. Target relationship research helps to better accomplish the cognitive task of mapping images to subtitles [22], and can also help improve image search tasks [23]. In this paper, we take visual relationship prediction as a pre-requisite, and use the principle of word vector similarity and relational predicates to help the QA model to improve the accuracy of the answer.

VQA research method: Malinowski et al. [24] designed a closed VQA system. The answer comes from pre-designed 16 basic colors and 894 objects, which can handle complex problems in reality. Tu et al. [25] jointly parsed the video and the corresponding text content and tested it on two data sets containing 15 video samples. In real situations, such as helping the visually impaired, questions and answers are open-ended. Open-ended visual questions may be directed to any area, any target, background details, and the relationship between targets of the image. Therefore, a successful VQA system usually requires a more detailed understanding of the image and complex reasoning than a system that generates generic image subtitles. Agrawal et al. [26] proposed a free-form open-ended visual VQA model. The model can provide accurate natural language answers by entering images into the model and relevant natural language questions. Mengye Ren et al. [27] considered that VQA system lacks criteria for judging the accuracy of sentences, and proposed to use a word as the answer to the visual VQA system, transforming the VQA task into a multi-category problem, and evaluating the answer with existing accuracy evaluation criteria. Mateusz et al. [28] combined the CNN model with the LSTM model to construct a VQA model with a variable answer length. The model interprets the VQA task as a process of generating image data using an auxiliary natural language sequence. Xu et al. [29] increased the computational efficiency of the VQA model by introducing attention mechanism to increase the weight of specific parts (images and/or questions) in the input vector. The main idea of the attention mechanism is to replace the overall (image range) features with spatial feature maps and to allow interaction between the question and specific areas of these maps.

3. Visual relationship prediction model

The basic structure of the VQA model is to directly extract image information with CNN, and then send the image features into LSTM to produce prediction results. In this paper, the target combination feature vector and the target relationship predicate generated by the image appearance relation model are used to provide image information for the image. The image appearance model consists of two parts: target detection model and target relationship judgment model.

3.1. Target detection model

When describing the image content with a sentence, the target in the image should be firstly detected for attribute labels that will become the subject or object in the description statement. Therefore, it is imperative that the deep convolutional nerve network model for target detection should be firstly trained. We referred to the method proposed

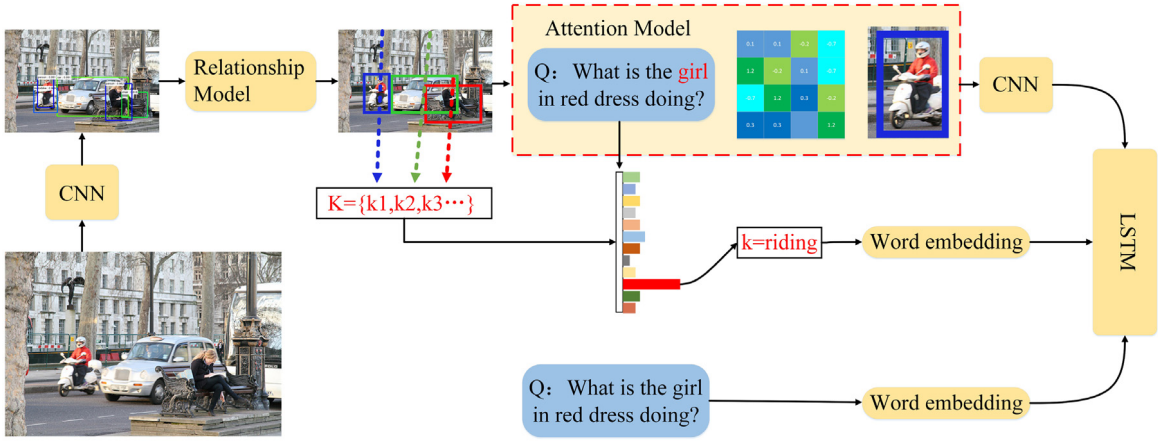


Fig. 1. Image Question Answering Model based on Visual Relationship Detection.

Fig. 2. Visual relationship detection: the given image serves as input, and we examine multiple relationships in the form of target 1-relation predicate-target 2.

by Karpathy et al. [30] and used Faster Region Convolutional Neural Network (Faster R-CNN) to detect the targets contained in the image. In order to improve the detection accuracy of the model, we selected the ImageNet [31] data set to pre-train the CNN, and the detection category is N = 200. ImageNet is a large visualized database for visual target recognition, with more than 14 million images manually annotated, and at least 1 million images are given the bounding box. After completing the target detection, we selected the top 20 targets and recorded the location of the bounding box containing those targets.

3.2. Target relationship judgment model

The task of the target relationship judgment model is to locate the target position in the image that is associated with the VQA system and give the corresponding relationship prediction. In natural language processing, the target relationship is defined as $\langle subject\text{–}predicate\text{–}object\rangle$ [32]. In order to facilitate understanding, this paper regards the visual relationship as a pair of local targets connected by relational predicates, and its relational structure is $\langle object1\text{–}relationalpredicate\text{–}object2\rangle$.

As shown in Fig. 2, this relationship structure is very common in reality. We detected the following relationships: $\langle person\text{–}ride\text{–}horse\rangle$, $\langle dog\text{–}beside\text{–}horse\rangle$ and $\langle person\text{–}wear\text{–}jacket\rangle$. In general, the distance between the bounding boxes of the two targets in the same relational structure will be very close or even overlap, and therefore the two target areas in the same relational structure can be combined. As shown in Fig. 3, the bounding box containing the human and the bounding box containing the horse can be combined into an overall bounding box, and then the target relational model is trained together with the corresponding relational predicate.

The relationship judgment model needs to train another CNN model to train the combined regions of the two participating targets and K = 50 relational predicates. In reference to Lu et al. [9], this paper defines the appearance relationship model V_R as follows:

$$V_R = P(O_1)(W_k^T CNN(O_1, O_2) + b_k)P(O_2) \tag{1}$$

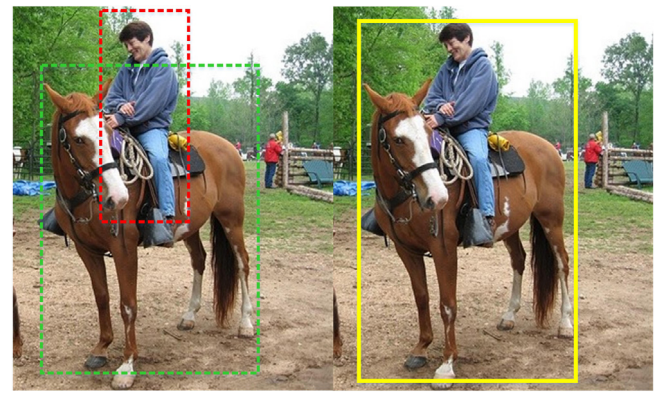


Fig. 3. Combine the appearance features of the two objects.

where, W_k and b_k are training parameters; k is the relationship predicate corresponding to the (O_1, O_2); $P(O)$ is the detection probability of the object; $CNN(O_1, O_2)$ is the joint feature information of the regions of O_1 and O_2.

4. Semantic relevance similarity detection

The VQA model can provide satisfactory answers to the questions in the training sets. However, when there is no relationship structure or target category in the training set, the VQA model has difficulty giving an accurate answer. In order to enhance the generalization of the VQA model, this paper further expands the appearance relationship model by the principle of word vector space similarity. By learning an appearance relationship structure, humans can give a relationship inference about the approximate structure. For example, even if we have not seen anyone riding a motorcycle, we can infer the relationship between people and motorcycles from a similar relationship structure that occurs frequently in life. Because people riding bicycles and people riding motorcycles have a similar appearance relationship structure. Therefore, the semantic relevance between the two targets can help the VQA model to make accurate relationship judgments.

Semantic relevance is mainly reflected in the closer distance between target vectors with similar relationship structure in word vector space. For example, $R_1 = (man\text{–}riding\text{–}cow)$, $R_2 = (people\text{–}riding\text{–}horse)$, and $R_3 = (girl\text{–}wearing\text{–}hat)$. We hope that in the word vector space, R_1 and R_2 are close but away from R_3. As the distance between two relations in the word vector space is proportional to the Euclidean distance between the transformed two relational structures [9], therefore the judging of the similarity of the target relationship can be transformed

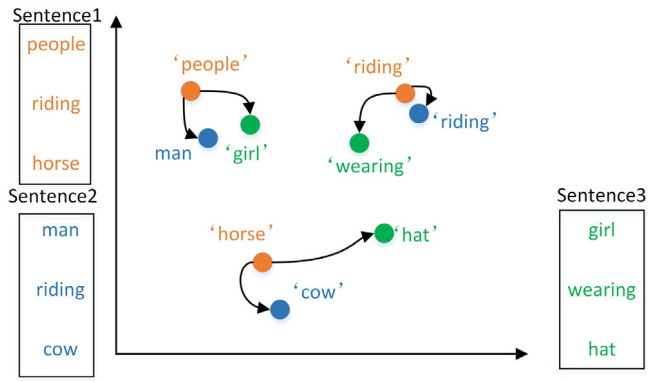


Fig. 4. Cost calculation of text transfer based on word matching.

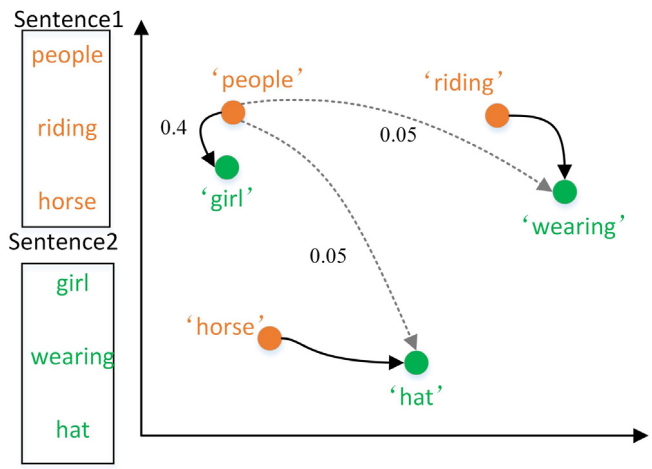


Fig. 5. Cost calculation of text transfer based on WMD.

into the judging of the similarity between the texts. To judge the similarity between two pieces of texts, the general method is to use the word2vec, a word embedding model, to convert the two pieces of texts into the word vector space, and then calculate the Euclidean or cosine distance between the two transformed structures, and finally compare the distance to determine the similarity of the two texts. Although this method is convenient for calculation, the relationship judgment only considers the distance of local words and ignores the global similarity studied in this paper. Therefore, when faced with complicated texts, it is easy to make judgment errors.

In this paper, the word mover's distance (WMD) algorithm [33] is used to calculate the similarity of word vectors. The WMD algorithm obtains the similarity between i and j (the semantic distance between i and j) by calculating the text migration cost. As shown in Fig. 4, the red boxes Sentence1, Sentence2, and Sentence3 are three sentences that need to be determined for relationship similarity. The direction of the arrow is from each word in sentence 1 to word that best matches them in sentence 2 and sentence 3. The general approach is to determine the relationship similarity of sentences by calculating the distance between the best matching pairs of words between sentences. In other words, the sizes of $d_{11}+d_{12}+d_{13}$ and $d_{21}+d_{22}+d_{23}$ in Fig. 4 is compared to determine which (Sentence2 or Sentence3) is more similar to Sentence1. However, the disadvantage of this method is that the words used by humans are very rich in indication, and sometimes two sentences that are completely different in form might express very similar meanings.

In order to solve the above problem, it is necessary to add a weighting factor when calculating the word vector distance, which enables each word in Sentence1 to be matched with all the words in another sentence according to different weights. A sentence can be seen as a distribution of different words in accordance with grammatical rules. As shown in Fig. 5, the process of calculating the similarity between Sentence1 and Sentence2 can be seen as the migration cost required for words in Sentence1 to migrate to Sentence2. The single migration cost is equal to the product of the Euclidean distance and the weighting factor of the two words in the word vector space. For example, the word "people" has a weight of 0.5 in Sentence1, and we assign 0.4 to the weight from "people" to "girl", 0.05 to "wearing", and 0.05 to "hat". Therefore the migration cost is: Add two constraints at the same time:

(1) The sum of the weights of the words "people" to each word in Sentence2 should be equal to Weights in Sentence1.
(2) The weight of a word in Sentence2 is equal to the weight sum that is migrated to this word. For example, the weight of the word "wearing" in Sentence2 is equal to the weight from "people" to "wearing" + the weight from "riding" to "wearing" + the weight from "horse" to "wearing".

Through the above method, a total migration cost can be obtained, and then the migration cost can be minimized. To obtain the migration cost D of two sentences, we first calculate the transition matrix $T_{i,j}$ formed by the transition vector represented by the transition arrow in Fig. 5.

$$\sum_{j=1}^{n} T_{i,j} = d_i^a, \forall i, j \in 1, \dots, n \tag{2}$$

$$\sum_{i=1}^{n} T_{i,j} = d_j^b, \forall i, j \in 1, \dots, n \tag{3}$$

$$D = min \sum_{i,j=1}^{n} T_{i,j} C(i,j), T \geq 0 \tag{4}$$

where, d_i^a and d_j^b are the vectors composed of a and b; $C(i,j) = |vec(i) - vec(j)|$ is the Euclidean distance between i and j; n is the number of words in the sentence. Calculating the cost of migration between sentences can help judge the similarity between different sentences. When encountering a relationship structure that has not been learned in the training set, the similarity of the migration is selected as a reference sentence and the relationship predicate is obtained by comparing the similarity between the existing sentences in the training set.

5. Question embedding

The VQA model in this paper consists of two parts: text question and the image appearance relationship feature. First, the text question is pre-processed. Each question is divided into words by spaces and punctuations, and words composed of numbers are also treated as a single independent word. Teney et al. [34] found that only about 0.25% of the question length was longer than 15 words after counting the length of the VQA question and answer data set. Therefore, in order to improve the computational efficiency, this paper divides a sentence by 15 words. After that, the words are transformed with $word2vec(*)$ into a 300-dimensional word vector. If the sentence is shorter than 15 words, it is filled with a zero vector. Finally, the embedding sequence consisting of the question sentence has a size of 15,300 and is embedded as a question in the vector into the LSTM network.

6. Question attention mechanism

Visual attention mechanism refers to a mechanism for processing brain signals, which is specific to human vision. Through quickly scanning the global image, human vision obtains the target area needing special emphasis, which is generally considered as the focus of attention. Then, more attention resources are invested on this area with the aim to obtain more details about the target while suppressing other useless information. The attention algorithms in this paper is different from other attention algorithms. The general method is to

divide the image into rectangular regions of equal size, and then use the attention algorithm to obtain the attention weight distribution of each rectangular region. In this paper, the feature extraction of the target appearance feature area is carried out, and the weight distribution of these different areas is calculated, which is more in line with human attention habits. Considering the characteristics of the VQA system, the question sentences are often closely related to the targets in the image. Therefore, in the process of generating the answers, the words contained in the questions can be used to guide the attention to the corresponding regions in the image.

First, the question-directed attention is calculated. For the object regions $i = 1, 2, 3, \ldots, N$ in image, the feature vector v_i is directly related to the question embedding vector Q. The attention weight α associated with the region is obtained through the nonlinear layer f_a, and then a feature vector with attention weighting is obtained.

$$a_i = w_a f_a(v_i, Q) \tag{5}$$

$$\alpha = softmax(a) \tag{6}$$

$$V^{att} = \sum_{i=1}^{N} \alpha_i v_i \tag{7}$$

The w_a is obtained through training, and the $softmax(a)$ is used to normalize the attention weights and calculate the weighted sum of α and v_i, and finally the attention-grabbing V^{att} is obtained. The calculation of f is based on the method of Damien et al. [34].

$$y^* = tanh(Wx + b) \tag{8}$$

$$g = \sigma(W^* x + b^*) \tag{9}$$

$$y = y^* \circ g \tag{10}$$

7. Multimodal fusion generating answers

The obtained V^{att} and Q are converted to a uniform vector space by f. Similar to the method in [34], we use element multiplication to fuse the two vectors and obtain their joint embedding representation: E.

$$E = f_q(Q) \circ f_v(V^{att}) \tag{11}$$

VQA model uses the VQA v2 public data set for training, and each question has at least one answer. In the data set, each answer has an accuracy within [0,1], and therefore the VQA model can be considered as a multi-label classification. To classify the answers, the classification network is to be initialized. The last layer of the classification network is the fully connected layer. We add the answer features to each row in the weight matrix of the fully connected layer so that each candidate answer $answer_j$ corresponds to the j row in the linear mapping matrix Z. In order to initialize Z, the Glove transformation algorithm can be used to convert the candidate answer $answer_j$ into a word vector into the matrix Z. The final number is actually the joint embedding representation E of the image features and the question features and similarity between the nonlinear function f_o and the linear mapping matrix Z containing the answer features. The expression that defines the accuracy number s^* is as follows,

$$s^* = \sigma(Z \circ f_o(E)) \tag{12}$$

The σ is the activation function, which finally normalizes the number into [0,1] to predict the correctness of each candidate answer. This paper uses the cost function in [34] to optimize the model parameters.

$$C = -\sum_{i}^{M} \sum_{j}^{N} s_{ij} log(s_{ij}^*) - (1 - s_{ij}) log(1 - s_{ij}^*) \tag{13}$$

where M is the number of questions; N is the number of candidate answers; s_{ij} is the reference number; s_{ij}^* is the output number of the fully connected layer. As there is a wide variety of targets in reality, it is difficult for the relationship between various targets to be included in the VQA training set, which will result in an inaccurate match between the answers and the corresponding rows in the matrix. As shown in Fig. 6, the VQA model can first search for a similar relationship structure and obtain the subject in the relationship judgment model, and then give an approximate answer reference according to the principle of word vector similarity, and finally form the candidate answer according to the form of the subject–predicate.

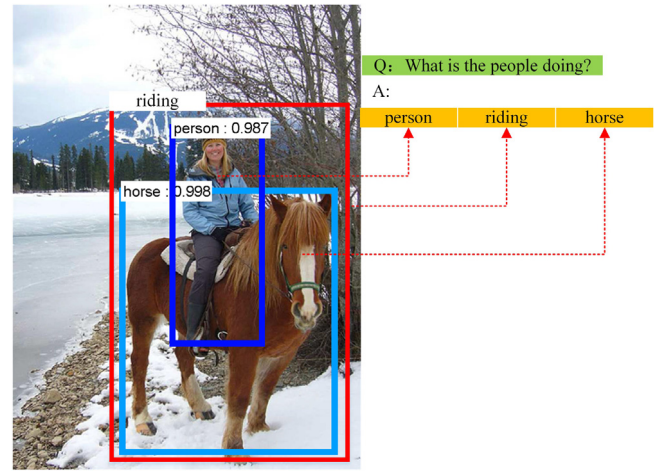


Fig. 6. Using object relationship detection to generate corresponding answers.

8. Experiments

8.1. Data set

The training database used in this paper can be classified into two categories according to their functions, and the two categories are aimed at training the question-answering model and the target relationship detection model, respectively.

DQAUAR (The Data set for Question Answering on Real-world images), the question-answering system database, is one of the earliest VQA training databases the smallest data sets. The images in DAQUAR are from the NYU-Depth v2 data set, which are all indoor RGBD images, with 795 pieces for training and 654 for testing. There are two major types of question/answer pairs in DAQUAR: automatically generated and manually labeled. VQA-v2 data set and VQA data set contain 123,287 training images and 81,434 test images. These images are mainly from the MS COCO data set. VQA contains 614,163 questions, and each question contains answers from different annotators.

Scene Graph data set, the target relationship detection database, contains 5000 scene images, 23,190 relationship types, and each object category is assigned 2–3 relational predicates. Visual Relationship database [9] contains 5000 scene images, 100 object categories and 70 predicates. The data set contains a total of 37,993 relationships, of which 6672 are relational types, and each object category contains 24–25 relational predicates. In the final model testing phase, 1000 images are selected from the Visual Relationship database for testing. The test set contains 1877 relationships that are never used in the training set. They are for verifying the generalization of the model when dealing with unknown relationship structures.

8.2. Quantitative analysis

In order to verify the effectiveness of the question-answering system model based on target relationship detection proposed in this chapter, the algorithm is compared with the current mainstream question-answering system algorithms based on multiple data sets. The performance of the question-answering model is evaluated and scored according to the Accuracy of the answer (Acc) [35] and Wu–Palmer

Table 1
Reported results on the DAQUAR-ALL data set.

DAQUAR-all	Acc.	WUPS@0.9	WUPS@0.0
Neural-Image-QA [37]	19.43	25.28	62.00
Multimodal-CNN [38]	23.40	25.59	62.95
Attributes-LSTM [39]	24.27	30.41	62.29
QAM [40]	25.37	31.35	65.89
Bayesian [41]	28.96	34.74	67.33
DPPnet [42]	28.98	34.80	67.81
ACK [43]	29.16	35.30	68.66
ACK-S [43]	29.23	35.37	68.72
SAN [44]	29.30	35.10	68.60
Our method	29.20	35.34	68.69

Table 2
Reported results on the DAQUAR-REDUCED data set.

DAQUAR-reduced	Acc.	WUPS@0.9	WUPS@0.0
GUESS [45]	18.24	29.65	77.59
Neural-Image-QA	34.68	40.76	79.54
Multimodal-CNN	39.66	44.86	83.06
Attributes-LSTM	40.07	45.43	82.67
QAM	42.76	47.62	83.04
Bayesian	45.17	49.74	85.13
DPPnet	44.48	49.56	83.95
ACK	45.79	51.53	83.91
ACK-S	46.13	51.83	83.95
SAN	45.50	50.20	83.60
Our method	45.77	51.56	83.90

Table 3
Reported results on the COCO-QA data set.

COCO-QA	Acc.	WUPS@0.9	WUPS@0.0
GUESS	6.65	17.42	73.44
Multimodal-CNN	54.95	65.36	88.58
Attributes-LSTM	61.38	71.15	91.58
QAM	58.10	68.44	89.85
Bayesian	63.18	73.14	91.32
DPPnet	61.19	70.84	90.61
ACK	69.73	77.14	92.50
ACK-S	70.98	78.35	92.87
SAN	61.60	71.60	90.90
Our method	69.67	77.08	92.14

similarity measure Set (WUPS) [36]. Acc consults image classification comparisons, for most of the answers are composed of single or multiple words, and the algorithm effectiveness can be assessed basically by the accuracy of the words. WUPS can calculate the similarity based on the longest common subsequence of two words in the same classification tree. If the similarity between the predicted word and the standard answer is below the set threshold, the candidate answer score is 0, with 0.9 and 0.0 representing the WUPS thresholds. Tables 1–3 are the testal results of the algorithm based on DAQUAR-ALL, DAQUAR-REDUCED, and COCO-QA, respectively.

The testal results show that our algorithm is significantly better than most counterparts in terms of the three evaluation indicators: Acc, WUPS0.0 and WUPS0.9. In addition, For all the evaluated metrics ACK-S performs better than the proposed method. They design a visual question answering model that combines an internal representation of the content of an image with information extracted from a general knowledge base to answer abroad range of image-based questions. This essay focuses mainly on the problems relating with relations between targets, however, it could not offer direct help to results of quantitative analysis. The reason is that the question-guided attention mechanism introduced in this chapter can effectively prevent the model from focusing on the wrong image region. Moreover, the target relationship detection method based on word vector similarity can improve the robustness of the question-answering model. In addition, continuous fine-tuning of parameters and models can also effectively improve the algorithm. For example, the CNN and LSTM model parameters of the question-answering system are fine-tuned, and the target classification model and the relationship determination model are separately pre-trained.

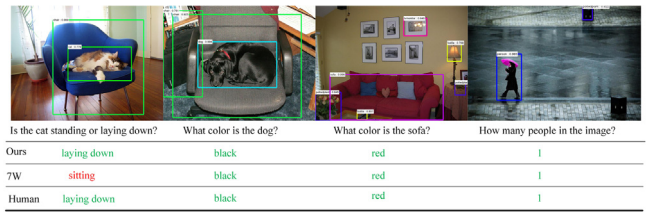


Fig. 7. Example of question and answer, Correct prediction is colored in green, incorrect in red.

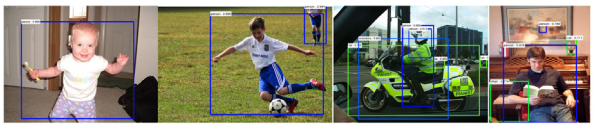
8.3. Qualitative analysis

In order to further verify the performance of the algorithm in reality, the question-answering system is tested based on different data sets in this chapter. The test is composed of question answering in simple and complex scenarios, simple and complex relationship detection, etc. In addition, corresponding target classification detection boxes are offered by all the test images, which are aimed at illustrating how questions guide the attention to find the answer target area among multiple classification detection areas when dealing with complex multi-objective situations. Further, in order to highlight the role of attention mechanisms and relationship detection in the question-answering system to generate answers, test results are compared with the 7 W algorithm [46–51] and the human benchmark answer. Test questions are answered with single words and word combinations. Based on the test results, although there are some erroneous answers by the algorithm in this chapter, the accuracy of the answer is improved compared with other algorithms, and proper speculations are made to handle some questions. When dealing with most question-involved scenarios in reality, the algorithm of this chapter can offer highly accurate answers. However, accurate answer generation is challenged by complex spatial structures, objects that are difficult to accurately identify, and uncertain target relationships.

Test 1: Questions on the pose, color, and quantity of objects in reality As shown in Fig. 7, the algorithm can offer an accurate answer when the object is stationary and unobstructed. For example, the question might be "What color is the sofa?" Although there are some objects with other colors on the sofa, the model can identify the main color of the sofa. In addition, in the training phase, the model can predict the scene structure by learning image scenes such as "the cat lies on the ground" and obtaining the target appearance relationship structure features.

Test 2: Questions on the identification, motion type, behavior identification of small objects in reality As shown in Fig. 8, target detection, especially the precise identification of small targets, is the basis for the question-answering system model to offer the correct answer. Illumination changes, occlusions, scale changes and other factors can affect the accurate identification of the target, and the wrong target semantic information passed to the question-answering system affects the output of the final answer. In addition, the language model lacks the ability to describe in detail. As shown in Fig. 8, the color of the police motorcycle is composed of multiple color stitches. As the model lacks targeted training, it is incapable of semantically describing complex graphics.

Test 3: Questions on image content, character behavior, color, and quantity in a multi-target scenario Tests 1 and 2 are about semantic question answering of a single target in simple images. In reality, there are multiple target types, and the spatial composition of different objects is different, and therefore the image content to be expressed is completely different. As shown in Fig. 9, the question is "what is the main content of the image", and the relevant image contains three main targets:



	What is on the right hand of the boy?	Which sport is this?	What color is the motorcycle?	What is this man doing?
Ours	toothbrush	football	green	man is reading
7W	stick	football	green	sitting
Human	toothbrush	football	green and white	the man is reading a book

Fig. 8. Example of question and answer, Correct prediction is colored in green, incorrect in red.

	What is the main content of the image?	What are they doing?	What color is the little girl's clothes?	How many people in the image?
Ours	a airplane is flying	the man is fighting	red	3
7W	airplane	playing the frisbee	red	3
Human	motorcycle cooperate with an airplane to perform stunts	they are wrestling	red	3

Fig. 9. Example of question and answer, Correct prediction is colored in green, incorrect in red.

	What is the woman in black coat doing?	What is he doing?	What is the man in red shirt doing?	What is he doing?
Ours	woman standing with snowboard	he is standing next to bike	he is riding	he is playing horizontal bar
7W	eating	talking on a bike	walking	using the signs
Human	a woman in black coat posing with a bear statue	he is repairing the wheel	he is riding horse	the man is exercising on the horizontal bars

Fig. 10. Example of question and answer, Correct prediction is colored in green, incorrect in red.

low-flying aircraft, motorcycles driving on the ground, and people climbing up through the rope ladder. Humans can accurately describe the behavior of each target monomer, and integrate the information to predict the theme to be expressed in the image, which is like a stunt performance. Traditional question-answering system models can only describe one of the many targets contained in the image as the answer. The algorithm in this chapter describes only one target and state.

Test 4: Questions on complex behavior in reality The question-answering system can be easily affected by object recognition accuracy and object ambiguity and other factors when dealing with various complex scenes in daily life. In particular, when dealing with questions that require a certain degree of logical reasoning rather than obtaining answers directly from the image, the question-answering system is too often stretched. As shown in Fig. 10, the question is "what is the woman in black coat doing", and the main target in the image is a woman wearing a black coat and a bear statue. The image is to describe a woman holding a bear statue. To generate the correct answer, the question-answering system has to correctly identify the two main targets in the image, especially to distinguish the "bear" from the "bear statue", and then to speculate that the theme is to describe a "group photo" rather than a "hug".

By analyzing the wrong answers generated in the above tests and based on actual application conditions, main factors making the model generate the wrong answers are as follows.

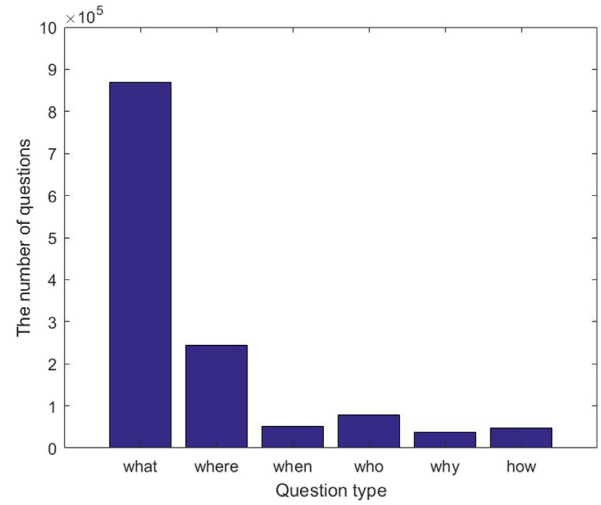


Fig. 11. Overall distribution of problem types in the data set.

(1) When dealing with image question answering in complex scenes, the model focuses on the wrong image area. When the subject describes the wrong object, the answer generated by the model must be wrong.
(2) Although the model focuses on the right area, the answer is ambiguous. Although sometimes the answer is different from the benchmark, it is also visually reasonable.
(3) When dealing with an ambiguous relationship structure or a target type that has not been in the training set, the question-answering system fails to understand the true theme the image is to describe, for it is incapable of logical reasoning.
(4) The data set label is markedly wrong, and the model can offer the correct answer, but the label is wrong.

9. Further discussion on question type

By analyzing the structured scenes for the question-answering system, image question answering based on image content alone is limited. Based on statistics on the images and questions of the question-answering system data set, only about 40% of the answers can be read by directly extracting the picture information, and the other 60% require logical reasoning and relationship prediction. Fig. 11 shows the statistics of different types of questions of the VQA-v2 data set.

The number of questions that start with "what" accounts for 56% of the total number of questions, and answers to "what" questions often involve information that does not correspond to a particular visual entity and therefore cannot be directly described by entities in the scene image. Because answers to "what" questions often require higher levels of abstraction, such as behavior, relationships, causes, weather, etc. Detecting the target in the current scene and abstracting the semantic information can help understand the content of the scene, but it is not comprehensive enough to capture all the elements needed for the answer generation. For example, underlying visual attributes (such as color and material), and the interdependencies between global scene attributes (such as location, weather, and time) and targets are hard to capture, but those information is essential for answering "what" questions.

10. Conclusion

In this paper, a question-answering system model based on target relationship detection is proposed to answer natural language questions about image content. Based on the attention-based image semantic description, the research incorporates a new attention mechanism and

theories relevant to word vector space to handle image semantic tasks of a higher level. The question-based attention mechanism can help the model focus on target areas most relevant to the question. Such a senior-concept-driven top-down attention mechanism is called concept-driven. Moreover, in this paper, target semantic information is converted into word vector space, and similar relationship structures are compared based on word vector similarity, thereby improving the generalization of the question-answering system model.

Acknowledgments

This work was supported in part by the Fundamental Research Funds for the Central Universities of China under Grant 2722019PY052 and by the open project from the State Key Laboratory for Novel Software Technology, Nanjing University, under Grant No. KFKT2019B17.

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